

# Modular Program Hierarchy

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## Hierarchy

- MealPlanner (class)
  - main() - program driver
    - askForBudget(), askForCalorieGoal(), askForMealType(), askForChoice() - all call askUser()
    - askForAnotherMeal() - Yes/No dialog
    - showMessage(), showError() - output dialogs
  - Menu arrays and constants - meal data used by main()

## Module Roles

| Module              | Purpose                                         | Returns                 |
|---------------------|-------------------------------------------------|-------------------------|
| main()              | Program driver: setup, meal loop, daily summary | None                    |
| askForBudget()      | Gets a positive daily budget                    | double budget           |
| askForCalorieGoal() | Gets gender (M/F) and sets calorie goal         | int goal (2000 or 2500) |
| askForMealType()    | Gets breakfast, lunch, or dinner (B/L/D)        | String meal type        |
| askForChoice()      | Shows the menu and gets a choice (1-3)          | int choice              |
| askForAnotherMeal() | Asks Yes/No for another meal                    | boolean                 |
| askUser()           | Shows a JOptionPane input box; exits on Cancel  | String answer           |
| showMessage()       | Shows a JOptionPane message box                 | None                    |
| showError()         | Shows a JOptionPane error box                   | None                    |

## Program Flow (main)

1. Show welcome message
2. Ask for budget and gender; set calorie goal (2000 female, 2500 male)
3. Loop: ask for meal type, show menu, get choice, check affordability
4. If affordable, subtract cost and add calories; show confirmation
5. End loop if out of budget or goal reached; otherwise ask for another meal
6. Show daily summary with jogging suggestion (if over) or success message